

Vector Search and Embeddings Deep Dive

Vector databases index high-dimensional vector representations of unstructured text data.

Chunking Strategies

Fixed-size Chunking: Splits text every N characters with fixed overlap.

Recursive Character Chunking: Respects natural structural boundaries such as paragraphs and headers.

Similarity Metrics

Cosine Similarity: Measures the angular distance between vectors, ideal for text embeddings.

Euclidean Distance: Measures straight-line spatial distance between points.

Query Pipeline Sequence

1. Embed the user query string using the sentence embedding model.
2. Query the vector collection for Top-K nearest neighbors.
3. Pass retrieved document chunks alongside the original user prompt to the local LLM.